Listing 4.8 `autogen_coding_critic.py`

```
from autogen import AssistantAgent, UserProxyAgent, config_list_from_json

config_list = config_list_from_json(env_or_file="OAI_CONFIG_LIST")

user_proxy = UserProxyAgent(
    "user",
    code_execution_config={
        "work_dir": "working",
        "use_docker": False,
        "last_n_messages": 1,
    },
    human_input_mode="ALWAYS",
    is_termination_msg=lambda x:
x.get("content", "").rstrip().endswith("TERMINATE"),
)

engineer = AssistantAgent(
    name="Engineer",
    llm_config={"config_list": config_list},
    system_message="""
    You are a profession Python engineer, known for your expertise in
software development.
    You use your skills to create software applications, tools, and
games that are both functional and efficient.
    Your preference is to write clean, well-structured code that is easy
to read and maintain.
    """,      #1
)

critic = AssistantAgent(
    name="Reviewer",
    llm_config={"config_list": config_list},
    system_message="""
    You are a code reviewer, known for your thoroughness and commitment
to standards.
    Your task is to scrutinize code content for any harmful or
substandard elements.
    You ensure that the code is secure, efficient, and adheres to best
practices.
    You will identify any issues or areas for improvement in the code
and output them as a list.
    """,      #2
)

def review_code(recipient, messages, sender, config):      #3
    return f"""
            Review and critque the following code.

            {recipient.chat_messages_for_summary(sender)[-1]['content']}
            """                         #3

user_proxy.register_nested_chats(      #4
    [
        {
            "recipient": critic,
            "message": review_code,
            "summary_method": "last_msg",
            "max_turns": 1,
        }
    ],
```